

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2021 - 2022)

JEE(Main)**UNIT TEST****21-11-2021****JEE(Main + Advanced) : NURTURE COURSE [PHASE : TEPS, TOPS, I(A), I, I(B), II(A), II & LIVE-I, I(B), II(A)]**

Time : 3 Hours

Maximum Marks : 300

READ THE INSTRUCTIONS CAREFULLY**Important Instructions :**

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The Test Booklet consists of **90** questions.
4. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
 - (i) Section-I contains 20 multiple choice questions with **only one correct** option.
Marking scheme : +4 for correct answer, 0 if not attempted and -1 in all other cases.
 - (ii) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
Marking scheme : +4 for correct answer and 0 in all other cases.
5. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited.**
6. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
7. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
8. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
9. **Do not fold or make any stray marks on the Answer Sheet.**
10. Take $g = 10 \text{ m/s}^2$ unless otherwise stated.

DO NOT BREAK THESE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Name of the Candidate (in Capitals)_____

Form Number : in figures_____

: in words_____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

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PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

1. The dimensions of four wires of the same material are given below. In which wire increase in length will be maximum for a given force :

- (A) Length 100 cm, diameter 1 mm
(B) Length 200 cm, diameter 2 mm
(C) Length 300 cm, diameter 3 mm
(D) Length 50 cm, diameter 0.5 mm

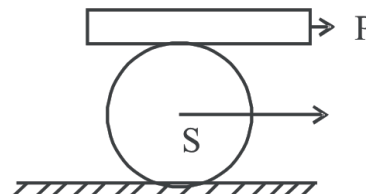
2. A wire elongates by ℓ mm when a load W is hanged from it. If the wire goes over a pulley and two weights W each are hung at the two ends, the elongation of the wire will be (in mm) :-

- (A) 0 (B) $\ell/2$
(C) ℓ (D) 2ℓ

3. A fixed volume of iron is drawn into a wire of length L . The extension x produced in this wire by a constant force F is proportional to

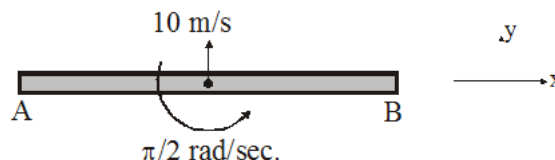
- (A) $1/L^2$ (B) $1/L$
(C) L^2 (D) L

4. A plank 'P' is placed on a solid cylinder S which rolls on a horizontal surface. The two are of equal mass. There is no slipping at any of the surfaces in contact. The ratio of kinetic energy of 'P' to the kinetic energy of 'S' is :



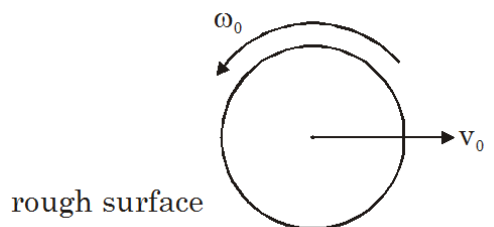
- (A) 1 : 1 (B) 12 : 1
(C) 8 : 3 (D) 11 : 8

5. A uniform rod AB of length 4 m and mass 12 kg is thrown such that just after the projection the centre of mass of the rod moves vertically upwards with a velocity 10 m/s and at the same time it is rotating with an angular velocity $\frac{\pi}{2}$ rad/sec about a horizontal axis passing through its mid point. Just after the rod is thrown it is horizontal and as shown in the figure. The acceleration (in m/s^2) of the point A when the centre of mass is at the highest point is (Take : $\pi^2 = 10$)

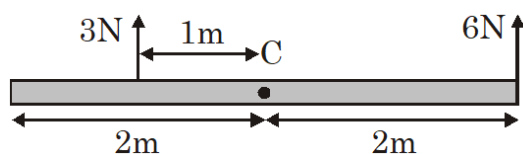


- (A) $5\hat{j}$
(B) $-10\hat{j}$
(C) $-5\hat{j}$
(D) $5\hat{i} - 10\hat{j}$

6. A uniform disc of radius $R = 0.2 \text{ m}$ kept over a rough horizontal surface is given velocity v_0 and angular velocity ω_0 . After some time its kinetic energy becomes zero. If $v_0 = 10 \text{ m/s}$, find ω_0 .



- (A) 50 rad/sec.
(B) 75 rad/sec.
(C) 100 rad/sec.
(D) 125 rad/sec.
7. A rod of mass 1 kg is kept on frictionless table and is acted upon by 2 forces as shown. How far from the center C of the rod should we apply a single equivalent force which produces the same effect as produced by the two forces shown?

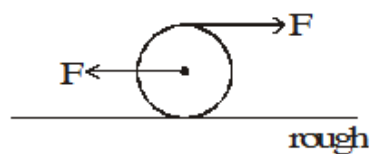


- (A) 1 m
(B) 1.1 m
(C) 1.2 m
(D) 1.5 m

8. A constant force acting at the centre of a uniform disc of radius r is always perpendicular to the plane of the disc. The disc can rotate about a chord at a distance x from the centre of the disc. For what value of x will the angular acceleration of the disc be maximum :-

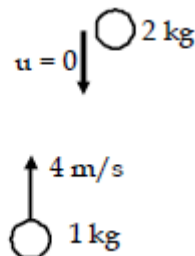
- (A) $\frac{r}{\sqrt{3}}$ (B) $\frac{r}{3}$
(C) $\frac{r}{2}$ (D) $\frac{r}{\sqrt{2}}$

9. What should be direction of friction if object pure roll on surface.



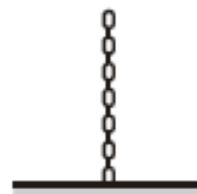
- (A) Rightward
(B) Leftward
(C) Friction does not act
(D) none of these
10. A ball of mass $m = 1 \text{ kg}$, travelling with velocity $\hat{i} - 3\hat{j}$ receives an impulse of magnitude 2 N-s in the negative x -direction. What is the velocity of the ball immediately after the impulse?
- (A) $-\hat{i} + \hat{j}$
(B) $-\hat{i} + 3\hat{j}$
(C) $-\hat{i} - 3\hat{j}$
(D) $\hat{i} - 3\hat{j}$

11. A ball of mass 1 kg is thrown up with an initial speed of 4 m/s. A second ball of mass 2 kg is released from rest from some height as shown in the figure. Select the correct alternative.



- (A) The centre of mass of the two balls comes down with acceleration $g/3$.
 (B) The centre of mass first moves up and then comes down.
 (C) The acceleration of the centre of mass is first g downwards and then g upwards.
 (D) The centre of mass of the two balls remains stationary.
12. A particle is moving with constant velocity has initial momentum p is given an impulse of magnitude I , If there is no change in its kinetic energy then
- (A) angle between its initial momentum and impulse must be $< 90^\circ$
 (B) angle between its initial momentum and impulse must be $> 90^\circ$
 (C) angle between its initial momentum & impulse is 90°
 (D) not possible

13. A uniform chain of mass m and length ℓ hangs on a thread and touches the surface of a table by its lower end. Thread breaks suddenly. Find the force exerted by the table on the chain when half of its length has fallen on the table. The fallen part does not form heap.



- (A) $\frac{mg}{2}$ (B) $\frac{3mg}{2}$
 (C) mg (D) none of these
14. A water fall is 84 metres high. If half of the potential energy of the falling water gets converted to heat, the rise in temperature of water will be
- (A) 0.098°C (B) 0.98°C
 (C) 9.8°C (D) 0.0099°C
15. The coefficients of thermal expansion of steel and a metal X are respectively 12×10^{-6} and 2×10^{-6} per $^\circ\text{C}$. At 40°C , the side of a cube of metal X was measured using a steel vernier callipers. The reading was 100 mm. Assuming that the calibration of the vernier was done at 0°C , then the actual length of the side of the cube at 0°C will be
- (A) > 100 mm
 (B) < 100 mm
 (C) $= 100$ mm
 (D) data insufficient to conclude

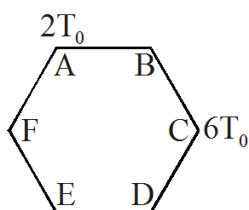
16. A metallic rod is continuously heated at its two ends, the flow of heat through the rod does not depend upon :

(A) The area of cross-section of the rod
(B) The mass of the rod
(C) Time
(D) The temperature gradient

17. One kilogram of ice at 0°C is mixed with one kilogram of water at 80°C . The final temperature of the mixture is (Take specific heat of water = $4200 \text{ kJ/kg-}^\circ\text{C}$, Latent heat of ice = 336 kJ/kg)

(A) 0°C (B) 40°C
(C) 50°C (D) 60°C

18. Six identical conducting rods having uniform cross-section area are fixed to make a regular hexagon as shown. The ends A and C are maintained at constant temperature $2T_0$ and $6T_0$. After the steady state is reached, pick up the correct relation between temperatures T_B , T_D , T_E and T_F of ends B, D, E and F respectively : (Neglect convection and radiation losses)



(A) $T_D > T_E > T_F = T_B$
(B) $T_D = T_B > T_E > T_F$
(C) $T_F < T_B = T_E < T_D$
(D) $T_B > T_F = T_E > T_D$

19. The natural length of a metallic rod at 0°C is l_1 , at $\theta^\circ\text{C}$ is l_2 . The given length of the rod at $\theta^\circ\text{C}$ is l_3 , then choose the incorrect options :

(A) The thermal stress in the rod is non zero if $l_3 > l_2$
(B) The strain in the rod is zero if $l_2 \neq l_3$
(C) The thermal stress in the rod is non-zero if $l_3 < l_2$
(D) The thermal stress and strain will be zero if $l_2 = l_3$

20. In natural convection a heated portion of a liquid moves because

(A) Its molecular motion becomes aligned
(B) Of molecular collisions within it
(C) Its density is less than that of the surrounding fluid
(D) Of currents of the surrounding fluid

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

- 1 kg mass of liquid A at temperature 30°C and 2kg mass of liquid B at 20°C are mixed in a calorimeter of negligible heat capacity. If final temperature of the mixture is 25°C . What will be the ratio of their specific heat capacity $\frac{S_A}{S_B}$?

- An aluminium vessel of mass 0.5 kg contains 0.2kg of water at 20°C . A block of iron of mass 0.2 kg at 100°C is gently put it to water. If the equilibrium temperature of mixture is $T^\circ\text{C}$. Fill the value of $\frac{T}{5}$ in OMR.

$$S_{\text{iron}} = 470 \text{ J kg}^{-1}\text{K}^{-1}; S_{\text{Al}} = 1140 \text{ J kg}^{-1}\text{K}^{-1};$$

$$S_{\text{water}} = 4200 \text{ J kg}^{-1}\text{K}^{-1}$$

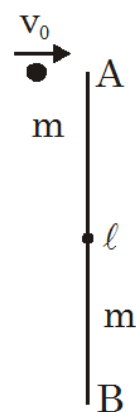
- A steel rod 25 cm long has a cross-sectional area of 0.8 cm^2 . Force required to stretch this rod by the same amount as the expansion produced by heating it through 10°C is:

(Coefficient of linear expansion of steel is $10^{-5}/^\circ\text{C}$ and Young's modulus of steel is $2 \times 10^{10} \text{ N/m}^2$.)

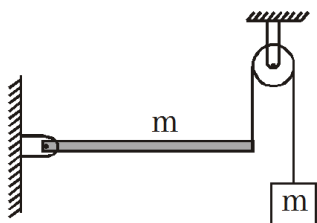
- A cylinder vessel of volume 1 litre having linear expansion coefficient of α is completely filled with a liquid of volumetric expansion coefficient γ . If temperature of vessel and liquid is increased by 50°C , find the volume of liquid spilled (in ml).

(Given $\alpha = 1 \times 10^{-5}/^\circ\text{C}$, $\gamma = 7 \times 10^{-5}/^\circ\text{C}$)

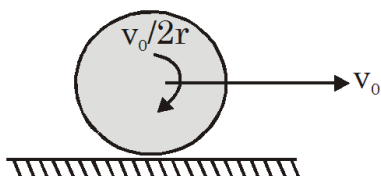
- A mass m is moving with speed v_0 perpendicularly to a uniform rod of mass m and length ℓ , which is initially at rest. The mass collides completely inelastically with the rod at one of its ends, and sticks to it as shown in the figure. The location of point from end A (in cm) which is at instantaneous rest just after collision is x_0 . The value of $x_0/10$ is (take $\ell = 1.2\text{meter}$)



6. A rod and a block are same mass. The rod is hinged at the other end. Initially rod is in horizontal position. When system is released from this position, the acceleration of tip of the rod is $\frac{3g}{K}$. Find the value of K.

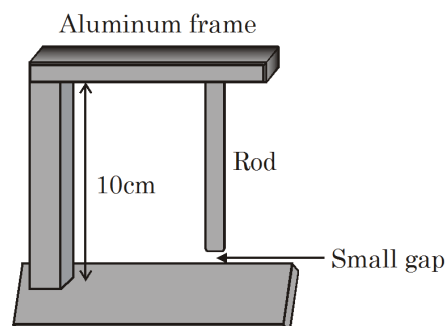


7. A sphere of mass M and radius r slips on a rough horizontal plane. At some instant it has translational velocity v_0 and rotational velocity about the centre $\frac{v_0}{2r}$. The translational velocity when the sphere starts pure rolling motion found to be $\frac{\beta v_0}{7}$. Find the value of β .

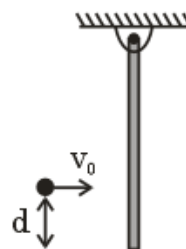


8. A 200 g, 20-cm diameter plastic disk is spun on an axle through its center by an electric motor. What torque (in N-m) must the motor supply to take the disk from 0 to 1800 rpm in 4.0s ?
(Take : $\pi = 3.14$)

9. At a certain temperature, a rod is hung from an aluminum frame, as the drawing shows. A small gap of 1mm exists between the rod and the floor. The frame and rod are heated uniformly. At what temperature change (in $^{\circ}\text{C}$) will the rod touch the floor, assuming that the rod is made from lead ? The coefficient of thermal expansion of aluminum is $23 \times 10^{-6} / ^{\circ}\text{C}$ and for lead it is $29 \times 10^{-6} / ^{\circ}\text{C}$. The height of the frame is 10 cm.



10. A ball of mass m is moving with speed v_0 . It undergoes a perfectly elastic collision with a rod of length L and mass M. The rod is pivoted about a frictionless axle through its edge and initially it hangs straight down and is at rest. What is the distance d such that the impulse applied by hinge is zero during collision. Neglect gravity. (Take : $m = 5 \text{ kg}$, $M = 20 \text{ kg}$, $L = 10 \text{ m}$)



SECTION-I : (Maximum Marks: 80)

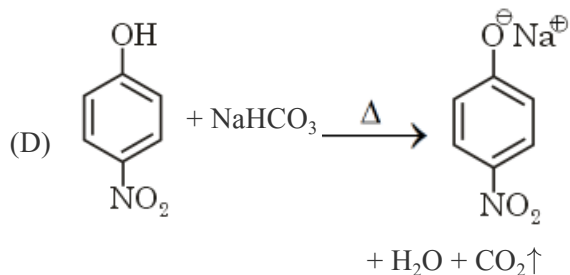
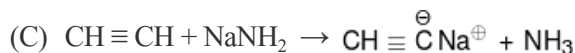
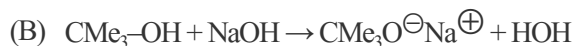
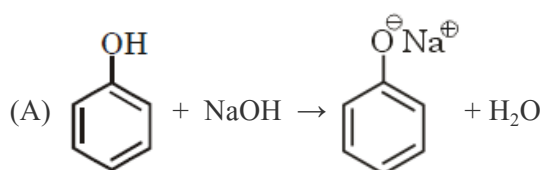
This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

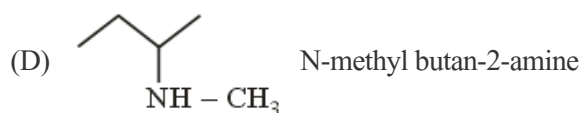
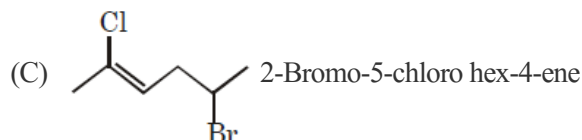
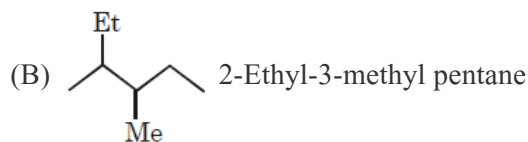
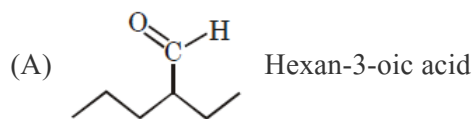
Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

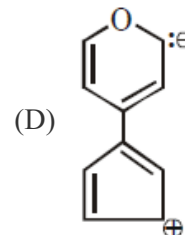
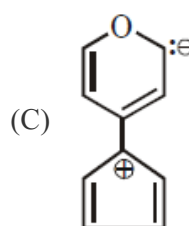
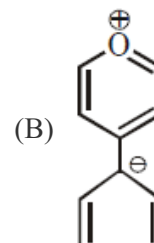
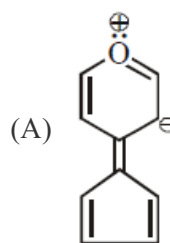
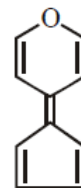
1. Identify reaction which is not feasible :



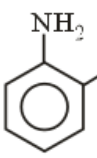
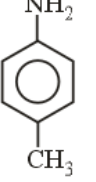
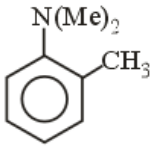
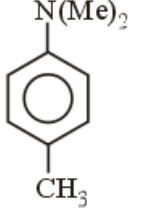
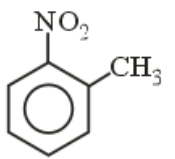
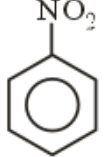
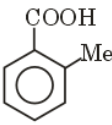
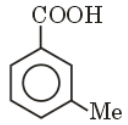
2. Which of the following is written with correct IUPAC name :



3. The most stable resonating structure for the given molecule :

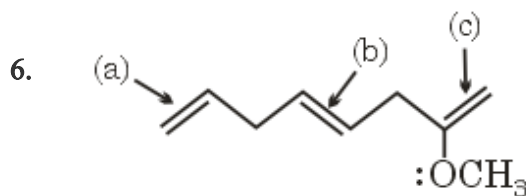


4. Select the correct statement ?

- (A)  is more basic than 
- (B)  is more basic than 
- (C)  has shorter C-N bond length than 
- (D)  is weaker acid than 

5. (I)  (II) 
 (III) 
 Correct option is

- (A) HOC order : I < II < III
 (B) HOH order : III < II < I
 (C) HOH per mole of H_2 : I > II > III
 (D) Stability order : III > I > II



Correct order of bond energy in above compound is :

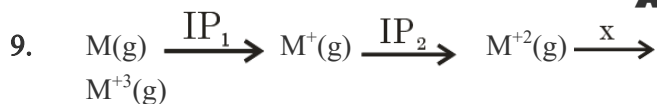
- (A) $c > b > a$ (B) $a > b > c$
 (C) $b > c > a$ (D) $b > a > c$

7. Which of the following order is CORRECT?

- (A) $\begin{array}{c} \text{CH}_3 \\ | \\ -\text{C}-\text{CH}_3 \\ | \\ \text{CH}_3 \end{array} > -\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_3 > -\text{CH}_2-\text{CH}_3 > -\text{CH}_3$
 (+I effect)
 (B) $-\overset{\oplus}{\text{N}}\text{H}_3 > -\overset{\oplus}{\text{N}}\text{Me}_3 > -\text{NO}_2 > -\text{CN}$
 (-I effect)
 (C) $-\text{Ph} > -\text{CH}=\text{CH}_2 > -\text{C}\equiv\text{CH}$ (-I effect)
 (D) $\text{F}^- > \text{Cl}^- > \text{Br}^- > \text{I}^-$ (Stability)

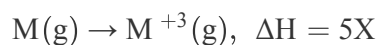
8. According to VSEPR theory, the shape $[\text{SF}_2\text{Cl}]^+$ and $[\text{S}_2\text{O}_4]^{2-}$ (around so called central atom) should be :-

- (A) Trigonal planar for $[\text{S}_2\text{O}_4]^{2-}$ and pyramidal for $[\text{SF}_2\text{Cl}]^+$
 (B) Both Trigonal planar
 (C) Pyramidal for $[\text{S}_2\text{O}_4]^{2-}$ and trigonal planar for $[\text{SF}_2\text{Cl}]^+$
 (D) Both pyramidal



The ratio $\frac{IP_1}{IP_2} = \frac{X}{2}$

Then calculate the value of IP_1 and IP_2 if



(A) $\frac{8X}{3+X}, \frac{4X}{2+X}$ (B) $\frac{8X}{2+X}, \frac{4X^2}{2+X}$

(C) $\frac{4X^2}{2+X}, \frac{8X}{2+X}$ (D) $\frac{4X}{2+X}, \frac{8X}{3+X}$

10. A planar sp^3 hybridised molecule having same surrounding atoms has 18% s-character in a bond. The % p-character in a lone pair of central atom is-

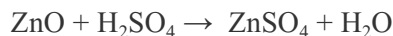
(A) 18% (B) 64%

(C) 82% (D) 68%

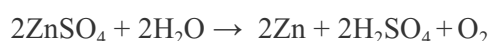
11. The chief ore of zinc is the sulfide, ZnS . The ore is concentrated by flotation and then heated in air, which converts the ZnS to ZnO .



The ZnO is then treated with dilute H_2SO_4



to produce an aqueous solution containing the zinc as $ZnSO_4$. An electric current is passed through the solution to produce the metal.



What mass of Zn is obtained from an ore containing 454 kg of ZnS ? Assume the flotation process to be 89.6% efficient, the electrolysis step to be 92.2% efficient, and the other steps to be 100% efficient. ($Zn = 65$)

(A) 252 gm (B) 252 kg

(C) 281.7 kg (D) 281.7 gm

12. An attempt was made to collect carbon dioxide, isolated from the decomposition of a carbonate containing mineral, by first bubbling the gas through pure liquid acetic acid. The experiment yielded 500 mL of a gaseous mixture of acetic acid and carbon dioxide at 1.00 atm and $16^\circ C$. The vapor pressure of pure acetic acid at $16^\circ C$ is 400 torr. What should be the total mass (approximately) of CO_2 and acetic acid vapours.

(A) 1.10 gm (B) 0.55 gm

(C) 2.2 gm (D) 11 gm

13. A mixture of a gaseous hydrocarbon and oxygen is in a vessel of a volume of 1 dm^3 at a temperature of 406.5 K and a pressure of 101 325 Pa. There is twice as much oxygen in the mixture as is needed for the reaction with the hydrocarbon. After combustion of the hydrocarbon the pressure in the vessel (at the same temperature) is increased by 5 %. What hydrocarbon was in the mixture when the mass of H_2O formed by the combustion was 0.162 g

$\left\{ \text{Assume : } 0.082 \times 406.5 = \frac{100}{3} \right\}$

(A) C_3H_4 (B) C_2H_6

(C) C_3H_8 (D) C_3H_6

14. The RMS velocity of H_2 at STP is $1.85 \times 10^5 \text{ cm/s}$ and its mean free path is $1.75 \times 10^{-5} \text{ cm}$. Find collision number (Use : $\sqrt{\pi} = 1.77$ if required)

(A) $9.7 \times 10^9 \text{ sec}^{-1}$

(B) $9.7 \times 10^{10} \text{ sec}^{-1}$

(C) $9.7 \times 10^8 \text{ sec}^{-1}$

(D) $9.7 \times 10^{11} \text{ sec}^{-1}$

15. Limonite is an ore of iron that contains $2\text{Fe}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$. A 0.5166 g sample of limonite is dissolved in acid and treated so that all the iron is converted to ferrous ions, Fe^{2+} . This sample requires 42.96 ml of 0.02130 M sodium dichromate solution, $\text{Na}_2\text{Cr}_2\text{O}_7$, for titration. Fe^{2+} is oxidized to Fe^{3+} , and $\text{Cr}_2\text{O}_7^{2-}$ is reduced to Cr^{3+} . What is the percent iron (by mass) in the limonite? ($\text{Fe} = 56$)

- (A) 14.87 %
(B) 1.487 %
(C) 5.95 %
(D) 59.5 %

16. A sample of crystalline soda (hydrated sodium carbonate) with a mass of 0.715 gm was decomposed by 50 cm³, 0.2 N H_2SO_4 . After total decomposition of soda, excess acid was neutralized by 50 cm³, 0.1 N KOH solution. Identify the correct statement -

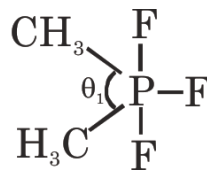
- (A) Soda is : $\text{Na}_2\text{CO}_3 \cdot \text{H}_2\text{O}$
(B) 5 millimole H_2SO_4 reacted with KOH
(C) Molar mass of soda is 250 gm
(D) Soda is $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$

17. An oxyacid of sulphur $\text{H}_2\text{S}_n\text{O}_6$ has total number of σ bonds = 11 then find the average oxidation state of sulphur :-

- (A) 3 (B) 2
(C) 2.5 (D) 4

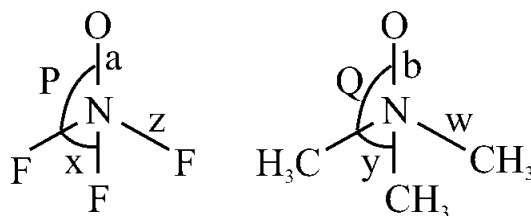
18. Select correct statement.

- (A) In tetrapolyphosphoric acid, four bridging oxo linkage are present.
(B) In Caro's acid, all oxygen atoms are in same oxidation state.
(C) Basicity of tetrathionic acid and pyrosulphuric acid is same.
(D) In tetrathionic acid all sulphur atoms having same oxidation state.

19. For the molecule  choose incorrect statement(s)

- (A) $d_{\text{P-C}} > d_{\text{P-F}}$
(B) $\theta_1 > 120^\circ$
(C) $d_{\text{P-F(axial)}} > d_{\text{P-F(equatorial)}}$
(D) All P-F bond length are identical

20. Which of the following options are **CORRECT**, about following observations.



- (A) $b < a$ (bond length)
(B) $P < Q$ (bond angle)
(C) $z < w$ (bond length)
(D) $x > y$ (bond angle)

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions. Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

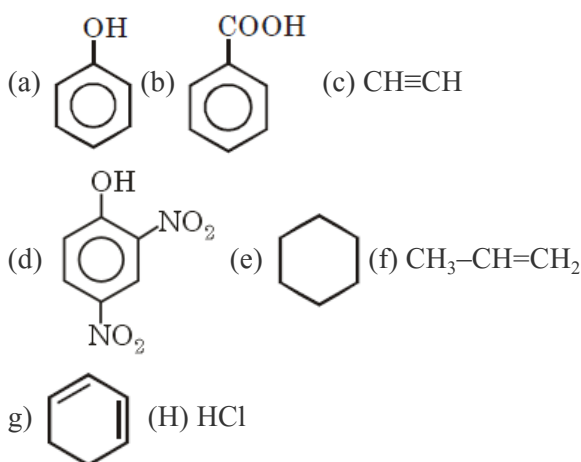
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

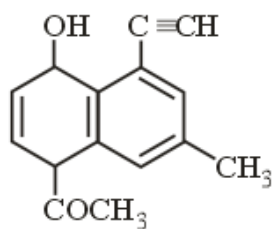
Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

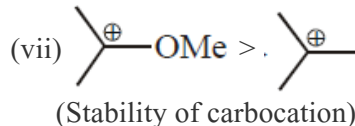
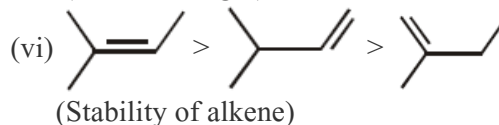
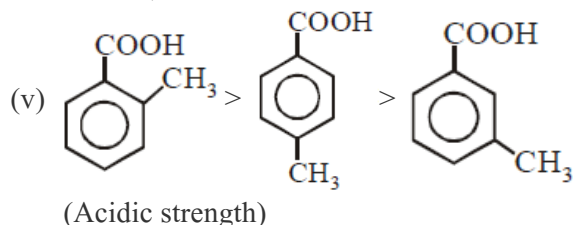
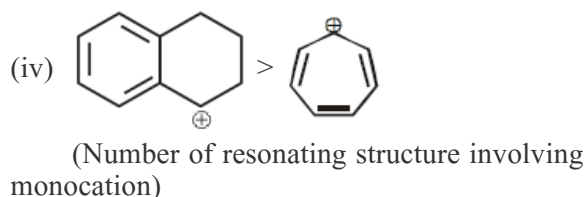
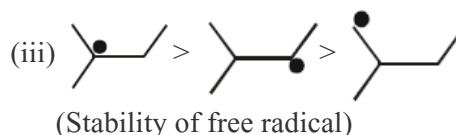
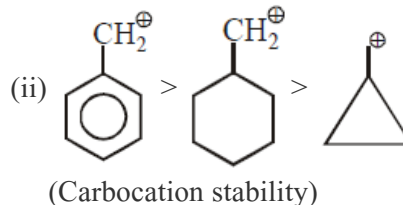
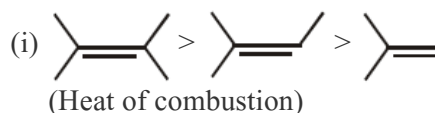
1. Number of molecules which will react with Na and NaOH both ?



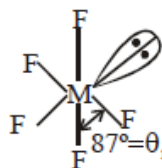
2. How many sp^2 carbons are present in following compound :



3. How many of the following orders are correctly matched ?



4. Structure of MF_5 is given below :

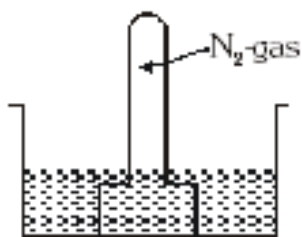


If $\text{F}-\text{M}-\text{F}$ (adjacent) bond angle in MF_4^- is θ_2
Find difference between θ_1 & θ_2
(M is same central atom in MF_5 and MF_4^-)

5. In how many following non polar species all adjacent bond angles are not same ?

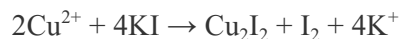
XeF_4 , PCl_5 , XeO_2F_4 , ClF_3 , C_2H_4 , SF_4 , IF_7

6. 190 mL of N_2 was collected in a jar over water at same temperature, water level inside and outside the jar standing at the same height (As shown in the fig.). If barometer reads 740 mm Hg and aqueous tension at the temperature of the experiment is 20 mm Hg, the volume (in mL) of the dry gas at 1 atm pressure and at the same temperature would be :

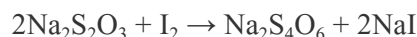


7. Two solutions of a substance (non electrolyte) are mixed in the following manner. 480 ml of 1.5 M first solution + 520 mL of 1.2 M second solution. What is the molarity of the final mixture ?

8. A 2.56 gm sample of copper ore is dissolved and $\text{Cu}^{2+}(\text{aq.})$ is treated with KI solution.



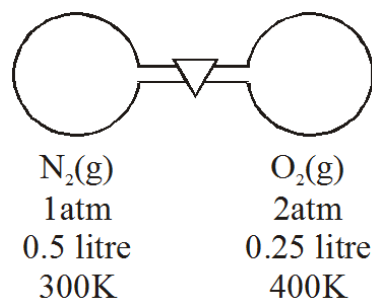
The liberated I_2 required 12.00 ml of 0.1M $\text{Na}_2\text{S}_2\text{O}_3$ solution for titration according to the reaction.



Calculate the % by wt. of Cu in ore

(Use : At. wt. of Cu = 64)

9. After opening stop cock, both bulbs are maintained at 400K. Final pressure in system at equilibrium will be (in atm)-



10. How many compounds have at least two peroxy linkage ($-\text{O}-\text{O}-$) ?

K_3CrO_8 ; Sodium peroxyborate ; CrO_5 ;

C_3O_2 ; $\text{H}_2\text{S}_2\text{O}_8$; H_2SO_5 , $\text{H}_4\text{P}_2\text{O}_7$

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

1. A (1, 3) and C $\left(-\frac{2}{5}, -\frac{2}{5}\right)$ are the vertices of a triangle ABC and the equation of the angle bisector of $\angle ABC$ is $x + y = 2$. Then equation of side BC is
 (A) $7x + 3y - 4 = 0$
 (B) $7x + 3y + 4 = 0$
 (C) $7x - 3y + 4 = 0$
 (D) $7x - 3y - 4 = 0$
2. Given A $\equiv$ (1, 1) and AB is any line through it cutting x-axis in B. If AC is perpendicular to AB and meets the y-axis in C, then equation of locus of mid-point P of BC is
 (A) $x + y = 1$
 (B) $x + y = 2$
 (C) $x + y = 2xy$
 (D) $2x + 2y = 1$

3. Let (1, 3) is centroid of an equilateral triangle whose one of side has equation $x + y - 5 = 0$, then equation of its circumcircle is -
 (A) $(x - 1)^2 + (y - 3)^2 = 4$
 (B) $(x + 1)^2 + (y + 3)^2 = 3$
 (C) $(x - 1)^2 + (y - 3)^2 = 2$
 (D) $(x - 1)^2 + (y - 3)^2 = \frac{3}{4}$
4. Pair of tangents are drawn from each point of line $x + y = 5$ to circle $x^2 + y^2 = 2$, then point of concurrency of their chords of contact, is -
 (A) $\left(\frac{2}{5}, \frac{2}{5}\right)$
 (B) (1, 1)
 (C) (-1, -1)
 (D) $\left(\frac{5}{2}, \frac{5}{2}\right)$
5. Angle between the tangents drawn to $\frac{x^2}{9} + \frac{y^2}{16} = 1$ from the point (3, 4) is -
 (A) $\frac{\pi}{3}$ (B) $\frac{\pi}{8}$
 (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{2}$
6. In the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$, the ratio of length of normal at P(having both coordinates non-zero) to the geometric mean of its focal radii is -
 (A) $\frac{3}{4}$ (B) $\frac{4}{3}$ (C) $\frac{3}{5}$ (D) $\frac{4}{5}$

7. Pair of tangents are drawn to $x^2 + 4y^2 + 4x + 3 = 0$ from points on the parabola $y^2 = 4x$. The corresponding chords of contact always touches a fixed conic section which is -
 (A) a circle
 (B) a parabola
 (C) an ellipse
 (D) a hyperbola
8. Let a variable chord PQ of parabola $y^2 = 8x$ always passes through point (6, 0). If OPRQ is a parallelogram, then locus of point R is (O being the origin) -
 (A) $y^2 = 4(x - 6)$
 (B) $y^2 = 8(x - 6)$
 (C) $y^2 = 4(x - 12)$
 (D) $y^2 = 8(x - 12)$
9. Length of focal chord $x - 2y = 2$ of parabola $y^2 = 8x$ is -
 (A) 40
 (B) $8\sqrt{5}$
 (C) 20
 (D) $4\sqrt{5}$
10. The locus of foot of perpendicular drawn from origin upon any tangent of parabola $y^2 = 8(x - 1)$ is
 (A) $x^3 + xy^2 + 2y^2 + x^2 = 0$
 (B) $x^3 - xy^2 + 2y^2 - x^2 = 0$
 (C) $x^3 - xy^2 + 2y^2 + x^2 = 0$
 (D) $x^3 + xy^2 + 2y^2 - x^2 = 0$
11. If the point P(4, -2) is the one end of the focal chord PQ of the parabola $y^2 = x$, then the slope of the tangent at Q is
 (A) $-\frac{1}{4}$ (B) $\frac{1}{4}$
 (C) 4 (D) -4
12. Slope of normal to the ellipse at a point P is $\frac{3}{4}$ and eccentricity of ellipse is $\frac{1}{3}$. If this normal makes acute angle ' β ' with its focal chord through P, then $\sin\beta$ is
 (A) $\frac{1}{4}$ (B) $\frac{3}{10}$
 (C) $\frac{1}{5}$ (D) $\frac{1}{2}$
13. The set of values of x in $(-\pi, \pi)$ satisfying $|4 \sin x - 1| < \sqrt{5}$ is given by -
 (A) $\left(-\frac{9\pi}{10}, \frac{-\pi}{5}\right)$
 (B) $\left(-\frac{\pi}{5}, \frac{\pi}{10}\right)$
 (C) $\left(-\frac{\pi}{10}, \frac{3\pi}{10}\right)$
 (D) $\left(\frac{3\pi}{10}, \frac{7\pi}{10}\right)$
14. The value of $\frac{\sin 5^\circ \sin 55^\circ \sin 65^\circ}{\sin 25^\circ \sin 35^\circ \sin 85^\circ}$ is -
 (A) $\sqrt{2} - 1$
 (B) $\sqrt{2} + 1$
 (C) $2 - \sqrt{3}$
 (D) $2 + \sqrt{3}$

15. If roots of the equation $ax^2 + bx + c = 0$ are α, β .

Then $\frac{1}{(\alpha^2 + \beta^2)} \left[\frac{1}{(a\alpha + b)^2} + \frac{1}{(a\beta + b)^2} \right]$ is

equal to -

- (A) a^2
 (B) c^2
 (C) $\frac{1}{c^2}$
 (D) $\frac{b^2 - 4ac}{2a^2}$

16. If a, b, c are in harmonic progression, then

$\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ are in -

- (A) A.P.
 (B) G.P.
 (C) H.P.
 (D) None of these

17. The number of integral values of k for which

the equation $3\sin x + 5\cos x = 2k + 1$ has a solution -

- (A) 5
 (B) 6
 (C) 7
 (D) 8

18. Let the area of $\triangle ABC$ is $\frac{\sqrt{3}-1}{2}$ sq. units, $b = 2$, and $c = (\sqrt{3}-1)$. If $\angle A$ is acute then the measure of angle C is

- (A) $\frac{5\pi}{12}$
 (B) $\frac{\pi}{3}$
 (C) $\frac{\pi}{6}$
 (D) $\frac{\pi}{12}$

19. The values of λ for which the system of equations

$$(\lambda + 5)x + (\lambda - 4)y + z = 0,$$

$$(\lambda - 2)x + (\lambda + 3)y + z = 0,$$

$$\lambda x + \lambda y + z = 0$$

has a non-trivial solution is/are -

- (A) $-1, 2$
 (B) $0, -1$
 (C) 0
 (D) No value of λ

20. The angle of elevation of the sun, if the length of the shadow of a tower is $\sqrt{3 - \sqrt{8}}$ times the height of the tower, is -

- (A) $\frac{3\pi}{8}$ (B) $\frac{5\pi}{12}$
 (C) $\frac{\pi}{12}$ (D) $\frac{\pi}{8}$

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

- Let x_1, x_2, x_3 are the solutions of the equation $x^{\log_2 x + (5\log_2 x) + 5} = \frac{4}{\left(\frac{1}{\sqrt{x+4}-2} - \frac{1}{\sqrt{x+4}+2}\right)}$, then $\frac{16}{25}(x_1 + x_2 + x_3)$ is
- With usual notations in a triangle if $3a = b + c$, then $\cot \frac{B}{2} \cot \frac{C}{2}$ is equal to
- If $\begin{vmatrix} x & y & -1 \\ y & -1 & x \\ -1 & x & y \end{vmatrix} = 0$, where $x, y \in \{-2, -1, 0, 1, 2\}$, then the number of ordered pairs of (x, y) is
- Number of points of intersection of the curves $3|x| + 4|y| = 12$ and $x^2 + y^2 = 16$ are

- From a point perpendicular tangents are drawn to the ellipse $x^2 + 2y^2 = 2$. The chord of contact touches a circle which is concentric with the given ellipse. Then the ratio of maximum and minimum area of circle is _____
- If $x + y = k$ is a normal to the parabola $y^2 = 12x$ and p is length of perpendicular from focus of parabola on this normal, then $\left(\frac{k}{3} + \frac{p}{\sqrt{2}}\right)$ is
- If A_1 be the A.M. and G_1, G_2 be two G.Ms between two positive number a and b , then $\frac{G_1^3 + G_2^3}{G_1 G_2 A_1}$ equal to
- In a triangle ABC using standard notation, $\frac{b^2 \sin 2C + c^2 \sin 2B}{\Delta}$ is always equal to -
- If α, β, γ are the roots of $4x^3 - 16x^2 + 19x - 6 = 0$, then the value of the determinant $\begin{vmatrix} \alpha - \beta - \gamma & 2\alpha & 2\alpha \\ 2\beta & \beta - \gamma - \alpha & 2\beta \\ 2\gamma & 2\gamma & \gamma - \alpha - \beta \end{vmatrix}$ is
- If $\frac{p}{q}$ is the minimum value of $\left|\frac{y}{x}\right|$ for pair of real number (x, y) which satisfy the inequality $x^2 + y^2 - 50y + 400 \leq 0$, then the value of $p + q$ is (p and q are relatively prime)

SPACE FOR ROUGH WORK